Principal Component Analysis (PCA)

input attribute

goal / target attribute →

	Whatsapp	Facebook	QQ	WeChat	Renren	IG	Discord	Mainland China
Ray	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Tony	Y	Y	Y	Y	N	Y	Y	N
Yukai	Y	N	Y	Y	N	Y	Y	Y
Ryan	Y	Y	Y	Y	N	Y	Y	N
Jory	Y	Y	Y	Y	N	Y	Y	Y
Donald	Y	N	Y	Y	N	N	Y	N
Ryan	Y	Y	N	N	N	Y	Y	N
X1	Y	Y	Y	Y	N	N	Y	Y
X2	Y	Y	Y	Y	N	N	Y	Y
Soya	Y	Y	N	*	N	Y	N	N
Eric	Y	N	N	Y	N	N	Y	N
Stara	Y	Y	Y	Y	N	Y	N	Y

if QQ = No,

Mailad Chaka = No

cor/. = 100%

if QQ = Yes,

Mailad Chaka = 50%

cut = 4/9

Similarity

- do data analysis

- Clustering : We do not have this "explicit" target attr.

- Classif° : We have an "explicit" target attr.

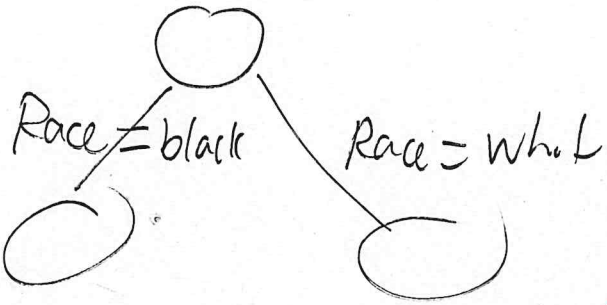
0 0 0 0 0	o_0
0 0 0 0 1	o_1
0 0 0 1 0	o_2
0 0 0 1 1	o_3
⋮	
1 1 1 1 1	o_{31}

1, 2, ... 8

entropy$_0$



Race



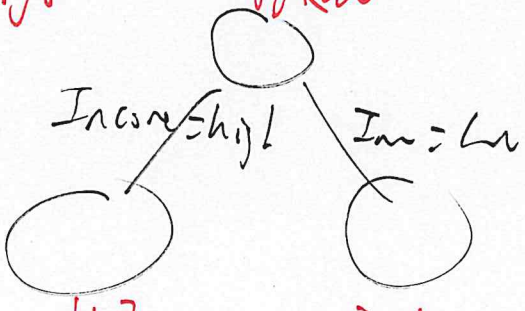
1, 5, 6, 7 — entropy$_{black}$

2, 3, 4, 8 — entropy$_{white}$

$$entropy_{Race} = \frac{4}{8} \times entropy_{black} + \frac{4}{8} \times entropy_{white}$$

$$Gain_{Race} = entropy_0 - entropy_{Race}$$

Income



1, 2 — entropy$_{high}$

3, 4, .. 8 — entropy$_{low}$

$$entropy_{Income} = \frac{2}{8} \times entropy_{high} + \frac{6}{8} \times entropy_{low}$$

$$Gain_{Income} = entropy_0 - entropy_{Income}$$

Child

$$entropy_{child} = \bigcirc \times \bigcirc + \bigcirc \times \bigcirc$$

$$Gain_{child} = entropy_0 - entropy_{child}$$

Ex 5 Q1

(a)

$$Info(T) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2} = 1$$

$$Info(T_{yes}) = -1\log 1 - 0\log 0 = 0$$

$$Info(T_{no}) = -\frac{1}{5}\log\frac{1}{5} - \frac{4}{5}\log\frac{4}{5} = 0.7219$$

$$Info(Child, T) = \frac{3}{8}Info(T_{yes}) + \frac{5}{8}Info(T_{no}) = 0.4512$$

$$Split\ Info(Child) = -\frac{3}{8}\log\frac{3}{8} - \frac{5}{8}\log\frac{5}{8} = 0.9544$$

$$Gain(Child, T) = \frac{Info(T) - Info(Child, T)}{Split\ Info(Child)} = \frac{1 - 0.4512}{0.9544} = 0.5750$$